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Professional Summary:

- **Senior Python Developer** with **10+ years** of experience designing and delivering scalable backend systems, distributed data platforms and API-driven architectures across finance, healthcare, telecom and retail. Proven success in end-to-end architecture, pipeline orchestration and multi-cloud deployments on AWS, Azure and GCP.
- Expert in **Python** application development using **Flask, Django, FastAPI, SQLAlchemy, Celery, Pandas** and **NumPy**, building modular, production-grade APIs, automation frameworks and high-performance services integrated with analytics, ML and third-party platforms.
- Specialized in distributed systems and real-time computation with **Apache Spark (PySpark), Apache Beam, Flink, Kafka, Kinesis, Azure Event Hubs, Google Pub/Sub, Apache Storm** and **Materialize**, enabling event-driven processing, fraud detection and streaming analytics.
- Strong background in data modeling, schema design and ETL/ELT logic using **Airflow, Celery, dbt, Pandas** and **NumPy**, ensuring scalable transformation architectures with built-in validation, testing and observability.
- Experienced in **relational** and **NoSQL databases** including PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, Cassandra, Spanner and Delta Lake, applying indexing, partitioning and query optimization for analytical and transactional workloads.
- Hands-on expertise in data warehousing using **Snowflake, Amazon Redshift, Redshift Spectrum, Google BigQuery** and **Azure Synapse**, applying star, snowflake and Data Vault 2.0 modeling for enterprise-scale analytics and reporting.
- Designed and orchestrated complex workflow automation using **Apache Airflow, AWS Glue, Azure Data Factory, Apache NiFi, Informatica, Talend, SSIS, Pentaho, Camel, Airbyte** and **Meltano**, ensuring modularity, maintainability and traceability across environments.
- Developed and deployed containerized Python services using **Docker** and **Kubernetes (EKS, AKS, GKE)**, managed via **AWS Fargate**, automated with Helm, ArgoCD, Argo Workflows and Tilt for fault-tolerant, CI/CD-driven scalability.
- Built and maintained CI/CD pipelines using **GitHub Actions, GitLab CI/CD, Jenkins, Azure DevOps, AWS CodePipeline** and **CircleCI**, integrating linting, unit tests, security scans and automated deployments.
- Proficient in Infrastructure-as-Code (IaC) frameworks including **Terraform, AWS CloudFormation, Azure ARM Templates, Google Deployment Manager, Pulumi** and **Azure Bicep**, ensuring reproducible and version-controlled environments for application and data systems.
- Implemented monitoring and observability using **Datadog, Prometheus, Amazon CloudWatch, Azure Monitor, New Relic, OpenTelemetry** and **Log Analytics**, establishing real-time dashboards, metrics and alerting pipelines.
- Experienced in MLOps and machine learning integration using **Amazon SageMaker, Google Vertex AI, Databricks, MLflow, Azure ML, Kubeflow** and **Spark MLlib**, building production-grade model inference APIs and data preparation pipelines.
- Applied AI/NLP and enrichment techniques using **SpaCy, NLTK** and **Hugging Face Transformers**, developing intelligent data pipelines for text classification, entity extraction and semantic search.
- Delivered visual analytics and embedded dashboards using **Matplotlib, Seaborn, Plotly, Altair, Power BI, Tableau, Looker, Amazon QuickSight** and **Google Data Studio**, integrating visual insights into backend reporting systems.
- Knowledgeable in data governance, lineage and metadata management using **Collibra, Alation, Apache Atlas** and **AWS Glue Data Catalog**, ensuring compliance, discoverability and traceability across enterprise environments.

Employment History:

Client: Truist Financial Corp , Charlotte,NC
Senior Python Developer

Jun 2023 – Till Date

Core Project Tasks:

- Architected and developed scalable, modular backend platforms using **FastAPI** and **Flask**, designing RESTful and GraphQL APIs that powered real-time analytics dashboards, risk-assessment systems and AI-driven financial compliance applications.
- Led architecture and requirement sessions to design **GCP**-based distributed systems, aligning **Python** microservices, **Redis** caching layers and **PostgreSQL** data-access strategies with product scalability, auditability and business goals.
- Engineered event-driven backend services using **Apache Kafka** and **Celery**, enabling asynchronous processing for fraud detection, payment clearing and transaction reconciliation pipelines.
- Designed workflow orchestration pipelines using **Apache Airflow** and **Celery Beat**, integrating **KYC/AML APIs** and credit scoring microservices into unified backend processing frameworks.
- Automated infrastructure provisioning with **Terraform** and **Google Cloud Deployment Manager**, implementing version-controlled **IaC** for production-grade financial systems.
- Developed optimized relational data models with **SQLAlchemy** and **PostgreSQL**, applying indexing, partitioning and constraint strategies to ensure ACID compliance and high availability.
- Built distributed processing modules with **PySpark**, accelerating compute-intensive portfolio analytics, liquidity forecasting and risk simulation workloads.
- Implemented streaming ingestion pipelines with asynchronous **Kafka consumers**, enabling near-real-time account updates, fraud alerts and transaction monitoring.
- Integrated data validation frameworks using **Great Expectations**, enforcing data integrity, lineage tracking and governance compliance across backend pipelines.
- Established distributed tracing and observability using **OpenTelemetry**, **Stackdriver** and structured logging, incorporating **AI-driven anomaly detection** for proactive performance monitoring.
- Orchestrated SLA-compliant job execution with **Celery**, **Redis Queue** and **Airflow DAGs**, ensuring timely execution of financial reporting and reconciliation workflows.
- Containerized applications using **Docker** and deployed to **Kubernetes (GKE)** with autoscaling and blue-green deployment strategies to ensure high availability and fault tolerance.
- Exposed secure production APIs via **FastAPI**, delivering aggregated financial analytics, **ML inference outputs** and **GenAI-enhanced insights** for internal compliance and reporting tools.
- Built CI/CD pipelines with **GitHub Actions**, automating testing, linting, containerization and secure release management workflows for production systems.
- Optimized backend performance by profiling request lifecycles, refactoring data-access patterns and tuning asynchronous I/O operations — reducing API response times by up to 35%.
- Integrated **TensorFlow Serving** with **FastAPI** for real-time credit scoring, anomaly detection and customer risk modeling APIs.
- Developed and maintained backend data-warehouse interfaces between **PostgreSQL** and **BigQuery**, implementing caching and read-replica strategies to scale analytical workloads.
- Built monitoring dashboards using **Grafana** and **Looker Studio**, visualizing API health, transaction latency and anomaly trends for operational intelligence.
- Implemented advanced security controls using **OAuth 2.0**, **JWT** and **Google IAM**, ensuring secure authentication, RBAC and compliance with **GDPR** and **PCI-DSS** standards.
- Engineered self-healing service mechanisms with retry strategies, idempotent APIs and fault-tolerant **Celery tasks** to improve system resilience and reduce operational incidents.
- Collaborated with **data science** and **DevOps** teams to design feature-serving APIs supporting ML-driven risk assessment and GenAI-powered decision support.
- Developed reusable data-preparation utilities using **Pandas** and **NumPy**, streamlining feature pipelines for real-time fraud detection and predictive analytics.
- Authored comprehensive API documentation and architecture standards using **OpenAPI**, ensuring maintainability, audit readiness and faster onboarding for new developers.
- Mentored junior developers through code reviews and training sessions on **Python** design patterns, asynchronous programming and CI/CD best practices, fostering technical excellence.

Client: UniversalHealthServices, Inc, King of Prussia, Pa
Senior Python Developer

Sep 2021 - May 2023

Core Project Tasks:

- Partnered with product and analytics teams to design and deliver scalable backend systems and data-access APIs using **FastAPI** and **Flask**, deploying modular Python microservices on **AWS Lambda**, **ECS** and **EC2** to orchestrate distributed compute workloads supporting oncology data pipelines and clinical analytics.
- Conducted architecture and code reviews across AWS-hosted applications to ensure SLA adherence, low-latency response and HIPAA-compliant data handling across multi-tier research platforms.
- Designed secure data storage and access architectures using **Amazon S3**, **RDS (PostgreSQL)** and **DynamoDB**, implementing **SQLAlchemy ORM** with advanced indexing and query optimization to accelerate assay-level transactional and analytical workloads.
- Automated infrastructure provisioning with **AWS CloudFormation** and **Terraform**, enabling reproducible, version-controlled and auditable deployments aligned with GxP-compliant infrastructure standards.
- Architected and managed **MongoDB Atlas** clusters integrated with backend APIs, applying sharding and replication to handle multi-terabyte genomics and immunoassay datasets with consistent throughput.
- Developed asynchronous and event-driven pipelines using **asyncio**, **AWS Kinesis Data Streams** and **Celery**, facilitating near real-time biomarker data synchronization, analytics updates and ML-driven research workflows.
- Engineered batch and on-demand data processing modules using **Pandas**, **NumPy** and **AWS Glue**, transforming structured and semi-structured datasets for bioinformatics modeling and pharmacokinetic simulations.
- Standardized API request validation and data serialization using **FastAPI** and **Pydantic**, enforcing schema and data integrity for clinical endpoints and internal analytical APIs.
- Developed modular data-access layers with **SQLAlchemy** integrating **PostgreSQL** and **Amazon Redshift**, optimizing query execution for multi-dimensional data marts supporting R&D analytics dashboards.
- Applied advanced schema design (normalized, denormalized and star-schema patterns) in **PostgreSQL** and **Redshift**, improving analytical performance for compound response analysis and trial data visualizations.
- Implemented scalable aggregation and transformation logic using **AWS Lambda** and **Python multiprocessing**, supporting machine learning feature extraction and AI-assisted drug response prediction pipelines.
- Established automated testing frameworks with **Pytest**, **unittest** and **Pydantic** validation suites, ensuring performance reliability and regulatory traceability for production APIs.
- Containerized backend services using **Docker** and deployed on **Amazon EKS (Kubernetes)**, achieving fault-tolerant and auto-scaled execution for microservices managing preclinical data workflows.
- Integrated **MLflow** into backend APIs to enable experiment tracking, model lineage and reproducible deployment of ML inference services across oncology analytics platforms.
- Designed observability frameworks using **Amazon CloudWatch**, structured JSON logging and custom **Python middleware**, incorporating AI-driven anomaly detection for proactive monitoring of compute pipelines and assay data flows.
- Developed CI/CD pipelines using **GitHub Actions** to automate testing, static analysis, container builds and multi-environment deployment into AWS staging and production environments.
- Integrated **Amazon SageMaker** inference endpoints into **FastAPI**, enabling real-time and batch biomarker prediction and clinical outcome classification models.
- Managed ML experiment metadata synchronization via **MLflow SDK**, maintaining traceability between deployed backend APIs and model artifacts across environments.
- Authored detailed technical documentation, data flow diagrams and API specifications using **Swagger** and **OpenAPI**, ensuring audit transparency, GxP compliance and streamlined developer onboarding.

Client: Fifth Third Bank, Cincinnati, OH
Python Developer

Jun 2019 - Aug 2021

Core Project Tasks:

- Led the redesign of enterprise Python backend services during a multi-cloud migration from **Azure** to **AWS**, re-architecting legacy ETL pipelines into modular, containerized **Flask** applications optimized for core banking data orchestration, compliance reporting and real-time account analytics.
- Developed high-throughput data ingestion and transformation modules using **Pandas**, **NumPy** and **Flask REST APIs**, enabling efficient movement of structured and semi-structured financial data across hybrid environments with secure PII masking.
- Built asynchronous, event-driven services using **Apache Kafka** and **Amazon Kinesis**, developing Python producers and consumers for streaming transaction telemetry and fraud analytics data with sub-second responsiveness.

- Designed and optimized data-access layers in **Amazon Redshift**, **PostgreSQL** and **Snowflake** using **SQLAlchemy**, implementing optimized indexing and partitioning for high-volume ledger, payment and customer analytics queries.
- Engineered distributed Python microservices using **Flask** and **Django REST Framework**, integrating with **Amazon SageMaker** for model-driven credit scoring, risk prediction and portfolio optimization endpoints.
- Managed complex workflow automation with **Celery** and **Apache Airflow**, orchestrating data ingestion, financial reconciliations and compliance validation tasks across production environments.
- Built secure RESTful APIs with **Flask**, implementing **OAuth 2.0**, **JWT** and **RBAC** mechanisms for authenticated access to internal banking systems and regulatory audit tools.
- Automated infrastructure provisioning and backend deployments using **Terraform**, **AWS CloudFormation** and **Azure Resource Manager**, embedding Python validation scripts to ensure cross-cloud configuration integrity and compliance traceability.
- Containerized backend applications with **Docker** and deployed on **Amazon EKS (Kubernetes)** and **Azure Kubernetes Service (AKS)**, ensuring fault-tolerant payment services through autoscaling, health checks and load balancing.
- Migrated user authentication and access management logic from **Azure AD** to **AWS IAM**, establishing encrypted, role-based access controls aligned with **PCI-DSS** and **FFIEC** standards.
- Implemented full-stack observability using **Prometheus** integrating **OpenTelemetry SDK** for distributed tracing and structured logging across financial microservices.
- Developed continuous integration and deployment pipelines using **GitLab CI/CD** automating testing, linting and environment-based releases with approval gates for regulated systems.
- Designed interactive data-visualization dashboards using **Matplotlib**, **Plotly** and **Amazon QuickSight**, embedding real-time transaction heatmaps and compliance trend insights into **Flask** admin interfaces.
- Built and maintained NLP enrichment modules using **SpaCy** and **NLTK**, converting unstructured text data such as loan documents, financial notes and transaction logs into structured insights for search and classification.
- Authored optimized **Python** and **Bash scripts** for data validation, log processing and automated audit trail generation, ensuring regulatory audit readiness under **SOX**, **AML** and **GLBA** frameworks.
- Developed cross-cloud asynchronous services using **AWS Lambda** and **Azure Functions** with custom Python encryption modules, enabling secure inter-system synchronization and encrypted data exchange between distributed services.
- Integrated **Fluentd** and **OpenTelemetry** for enhanced observability, trace correlation and anomaly detection in production APIs.

Client: Atlantic American Corp, Atlanta, GA

Oct 2017 - May 2019

Python Developer

Core Project Tasks:

- Designed and developed scalable, high-performance backend systems and batch-processing services using **Python**, **Pandas**, **NumPy** and multithreaded workflows, transforming terabyte-scale insurance claim and policy datasets for analytics and regulatory reporting.
- Built cloud-native backend services integrated with **PostgreSQL** and **Amazon RDS**, applying optimized schema design, indexing and query tuning to improve data performance for claims adjudication and actuarial analysis systems.
- Developed asynchronous, event-driven microservices using **Flask**, **asyncio** and **Apache Kafka**, enabling near real-time streaming of policy, premium and underwriting data across distributed systems.
- Implemented standardized transformation and validation logic through modular frameworks leveraging **SQLAlchemy** and custom schema enforcement classes, ensuring consistency and quality for financial and compliance datasets.
- Engineered reusable backend components and RESTful APIs with **Flask** and **Django**, integrating **AWS S3** storage for structured and semi-structured policy data ingestion and archival.
- Refactored inefficient SQL queries and backend access logic in **Oracle** and **PostgreSQL**, applying advanced indexing, connection pooling and ORM-level optimizations to enhance performance under peak data processing loads.
- Automated infrastructure provisioning and backend deployments using **Terraform** and **AWS CloudFormation**, ensuring consistent, version-controlled and auditable environments across staging and production.

- Containerized applications using **Docker** and deployed to **Kubernetes** clusters, utilizing autoscaling, liveness probes and resource quotas for fault-tolerant orchestration of insurance transaction services.
- Built and maintained RESTful APIs using **Flask** and **Django REST Framework**, integrating **JWT authentication**, caching and pagination to deliver secure, high-performance access to policyholder and claims APIs.
- Developed continuous integration pipelines using **Jenkins**, integrating **Pytest** and **flake8** and implementing rollback and static analysis controls to ensure reliable deployment of production services.
- Authored efficient **Python** and **Bash** scripts leveraging **awk** and **sed** for automated log parsing, system monitoring and audit validation, improving throughput across claims batch processing and regulatory compliance reports.
- Implemented observability using **Datadog SDK**, integrating metrics and structured logging into Python services to enable proactive detection of anomalies in data ingestion and ETL workflows.
- Integrated backend services with **Tableau** and **Power BI** APIs, automating dataset exports and embedding interactive risk and claims dashboards within **Flask**-based administrative portals.
- Defined software governance and coding standards, enforcing schema versioning, JSON contract consistency and unit-test coverage to meet **SOX** and **NAIC** insurance data compliance requirements.
- Collaborated with **data analysts**, **DevOps** and **actuarial teams** to design scalable APIs, reusable data-access layers and analytical microservices powering policy reporting, premium forecasting and financial data pipelines.

Client: Borg Warner INC, Auburn Hills, MI
Python Engineer

Sep 2015 - Sep 2017

Core Project Tasks:

- Developed automated data processing and backend workflows using **Python**, **Pandas** and **SQLAlchemy**, modernizing legacy ETL systems and improving performance and maintainability.
- Wrote and optimized **SQLAlchemy ORM** and **T-SQL** queries to support backend features in **PostgreSQL** and **SQL Server**, implementing stored procedures, indexed views and efficient joins for faster data retrieval.
- Built **Python integration scripts** and backend services to collect and validate data from **Oracle**, flat files (CSV/JSON) and **REST APIs**, standardizing structures for consistent downstream processing.
- Optimized application performance through query tuning, profiling and memory optimization in **Python** batch jobs, improving throughput and reducing execution times by up to 40%.
- Created lightweight **Flask** dashboards and web reports with **Plotly**, **Jinja2** and **Pandas**, allowing users to visualize KPIs and generate automated analytical reports.
- Cleaned and standardized unstructured datasets using **Python**, **Pandas** and **regular expressions**, applying validation and transformation logic to ensure accuracy and reliability of outputs.
- Collaborated with cross-functional teams to define API requirements and contribute to scalable **Python modules** following clean architecture and modular design principles.
- Designed normalized database schemas using **SQLAlchemy ORM**, applying indexing and relational modeling to improve query performance across analytical workloads.
- Used **Git** and **VS Code** for version-controlled development, participating in **code reviews**, **agile sprints** and collaborative issue tracking for continuous improvement.
- Automated recurring backend workflows with **Celery**, **cron jobs** and **Apache Airflow**, enabling reliable task scheduling and background job execution.
- Implemented structured logging, exception handling and retry mechanisms within **Python applications**, improving observability and resilience of automated processes.

Educational Background:

- Bachelors in Computer Science and Engineering.

Training & Certifications:

- AWS Certified AI Practitioner Foundational.
- Microsoft Azure Solutions Architect Associate.